DATA SCIENCE & EXPLORATORY DATA ANALYSIS | PYTHON

TITANIC SURVIVAL ANALYSIS

Exploratory Data Analysis & Statistical Modeling Report

Author: Sudipta Das **Domain: Machine Learning & Analytics** **Date: October 2026**

891	38.38%	74.20%	62.96%
TOTAL PASSENGERS	SURVIVAL RATE	FEMALE SURVIVAL	1ST CLASS SURVIVAL

1. Introduction and Objectives

The primary objective of this project is to explore the historical Titanic passenger dataset and understand how survival rates differed across passenger sex, passenger class, and age groups. Using Python for data handling, statistical exploration, and visualization, the study analyzes demographic patterns to identify key associations with survival outcomes.

2. Dataset Architecture and Tools

The dataset contains 891 passenger records and 12 original columns capturing survival status, ticket class, sex, age, fare, family relationships (SibSp, Parch), ticket details, cabin, and embarkation port. The analysis was conducted in Python using Pandas, NumPy, Matplotlib, and Seaborn.

3. Data Inspection and Preprocessing

A comprehensive data audit was conducted to assess structural integrity, summary statistics, and missing data. Missing values were detected across three attributes: Age (177 missing, filled with median), Cabin (687 missing, dropped due to severe >77% missingness), and Embarked (2 missing, filled with mode Southampton 'S'). The cleaned dataset contains 891 rows and 11 analytical columns with zero null values.

Attribute Name	Data Type	Preprocessing / Imputation Treatment
Age	Float	177 missing values filled with Median (robust against skewness)
Cabin	String / Code	687 missing values dropped entirely (>77% missing)
Embarked	Categorical	2 missing values filled with Mode (Southampton - S)
Survived, Pclass, Sex	Categorical / Numeric	Validated complete with zero missing values (100% integrity)

4. Exploratory Data Analysis (Population Overview)

Five visualizations were constructed to evaluate passenger survival dynamics. Figures 1 and 2 below illustrate overall survival distributions and passenger age demographics across the sample population:

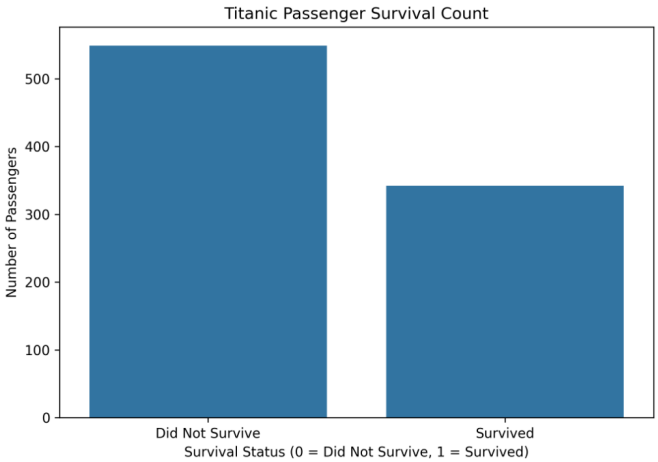


Figure 1: Passenger Survival Count

Out of 891 passengers, 342 survived and 549 did not survive, yielding an overall baseline survival rate of 38.38%.

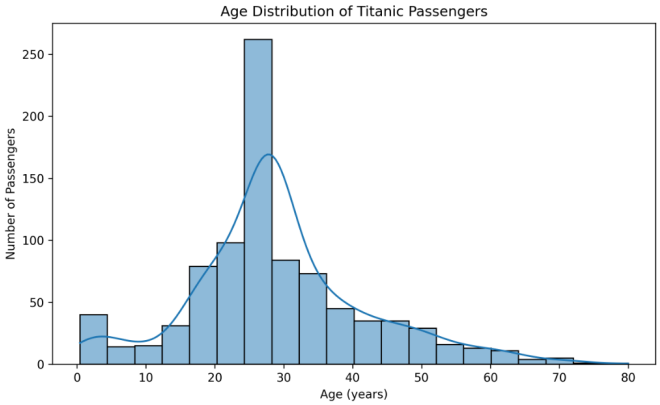


Figure 2: Age Distribution with KDE Curve

The histogram and density curve highlight that the majority of passengers were concentrated in their twenties.

Baseline Population Observations:

The baseline survival rate (38.38%) indicates that nearly two out of every three passengers perished during the disaster. The age profile is heavily concentrated between 20 and 35 years old, representing the large working-class demographic traveling in third-class accommodations.

4. Analysis and Visualizations (Continued - Sex, Class & Age)

Further disaggregation reveals strong survival stratification based on biological sex, ticket class tier, and developmental age categories:

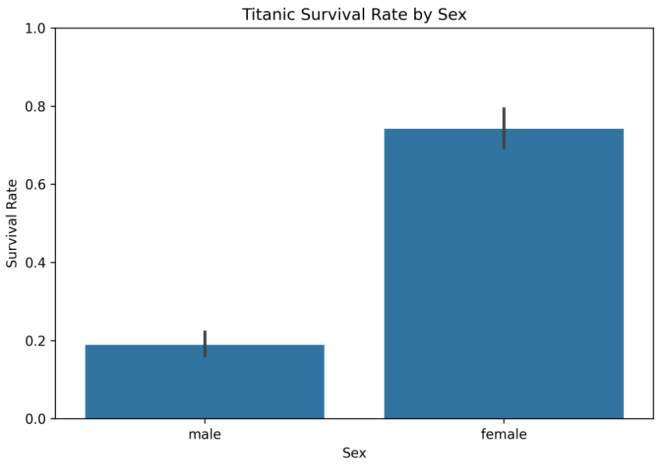


Figure 3: Survival Rate by Sex

Female passengers had a 74.20% survival rate vs 18.89% for males, verifying maritime evacuation protocols.

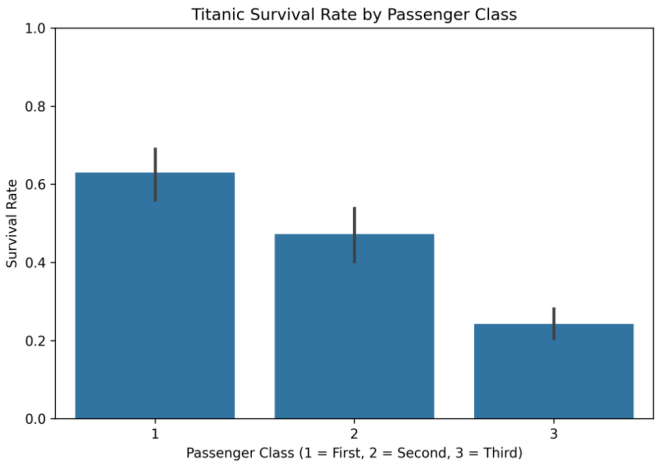


Figure 4: Survival Rate by Passenger Class

First class reached 62.96%, second class 47.28%, and third class 24.24%, showing direct socioeconomic stratification.

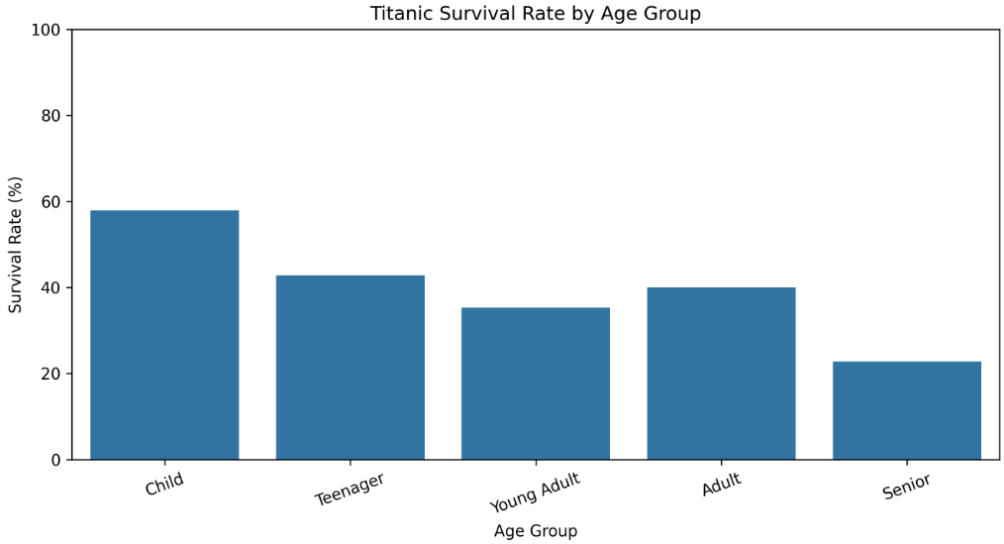


Figure 5: Titanic Survival Rate by Age Group

Children recorded the highest survival rate at 57.97%, followed by teenagers (42.86%), adults (40.00%), young adults (35.33%), and seniors lowest (22.73%), reflecting priority evacuation of vulnerable cohorts.

Quick Results Reference Matrix

Demographic / Operational Metric	Survived / Total	Survival Rate	Empirical Significance
Total Passengers Analyzed	342 / 891	**38.38%**	Overall baseline benchmark
Female Passengers	233 / 314	**74.20%**	~4x higher than male survival rate
Male Passengers	109 / 577	**18.89%**	Lowest survival among sexes
First-Class Passengers (Class 1)	136 / 216	**62.96%**	Highest class survival rate
Second-Class Passengers (Class 2)	87 / 184	**47.28%**	Intermediate class rate
Third-Class Passengers (Class 3)	119 / 491	**24.24%**	Steepest mortality rate
Children (Age 0 - 12)	40 / 69	**57.97%**	Highest cohort survival rate
Teenagers (Age 13 - 19)	30 / 70	**42.86%**	Above-average survival
Young Adults (Age 20 - 35)	181 / 512	**35.33%**	Largest population cohort
Adults (Age 36 - 55)	78 / 195	**40.00%**	Consistent with baseline
Seniors (Age 56+)	10 / 44	**22.73%**	Lowest cohort rate

5. Key Findings

- **Total Passengers:** A total of 891 passengers were included in the analysis.
- **Survival Counts:** 342 passengers survived, while 549 did not survive.
- **Overall Survival Rate:** The overall survival rate was 38.38%.
- **Gender Disparity:** Female passengers had a survival rate of 74.20%, compared with 18.89% for male passengers.
- **Class Disparity:** First-class passengers had the highest survival rate (62.96%), followed by second-class passengers (47.28%) and third-class passengers (24.24%).
- **Age Group Breakdown:** Children had the highest survival rate among the analyzed age groups (57.97%), while seniors had the lowest (22.73%).

6. Conclusion

This project uses Python to analyze the Titanic passenger dataset. The analysis examines overall survival, differences by sex and passenger class, and survival patterns across age groups.

The results help identify which passenger characteristics were associated with survival in this dataset. Specifically, female passengers, first-class passengers, and children exhibited significantly higher observed survival rates. This project reinforced essential data science competencies, including structured data inspection, missing value handling, cohort grouping, rate computations, and graphical communication.

7. Limitations

- **Age Imputation:** Some passenger ages were missing and were filled using the median age.
- **Cabin Feature Removal:** The Cabin column was removed because it contained many missing values.
- **Correlation vs. Causation:** The analysis identifies patterns and associations; it does not prove causation.
- **Methodological Sensitivity:** Missing-value handling and the selected age-group boundaries may affect the results.

8. Project Summary & Author Information

This document serves as the project report for the Titanic Survival Analysis project. All exploratory data analysis, data cleansing, statistical aggregations, and visual figures were developed using Python and standard scientific computing packages (Pandas, NumPy, Matplotlib, and Seaborn).

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Tech Stack: Python 3, Pandas, Seaborn

Project: Titanic Survival Analysis

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